

Rare Is Not Never

AP Statistics · Unit 2 · Topic 2.10 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **LO UNC-3.A** — Estimate probabilities of binomial random variables using data from a simulation. [Skill 3.A]
- **EK UNC-3.A.1** — A probability distribution can be constructed using the rules of probability or estimated with a simulation using random number generators.
- **EK UNC-3.A.2** — A binomial random variable, X , counts the number of successes in n repeated independent trials, each trial having two possible outcomes (success or failure), with the probability of success p and the probability of failure $1 - p$.
- **LO UNC-3.B** — Calculate probabilities for a binomial distribution. [Skill 3.A]
- **EK UNC-3.B.1** — The probability that a binomial random variable, X , has exactly x successes for n independent trials, when the probability of success is p , is calculated as $P(X = x) = C(n, x) \cdot p^x \cdot (1 - p)^{(n - x)}$, for $x = 0, 1, 2, \dots, n$. This is the binomial probability function.

Essential question: When does a low binomial probability count against a familiar chance model without proving it false?

DOK-3 skill: 4.B · **FRQ pattern:** binomial-conditions-then-is-this-outcome-unusual

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Uses 0.0106 to distinguish unusual from impossible and assess both claims
- **reasoning-2** (required) — Gives a plausible challenge to independence or constant p
- **reasoning-3** (required) — Names comparable new shooting evidence and explains what pattern would matter

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Computing only $P(X = 3)$ when the evidence is 3 or fewer
- Calling shots independent merely because there are 10 of them
- Treating a small probability as proof that the career probability changed
- Saying any result from 10 attempts must be ordinary because the sample is small

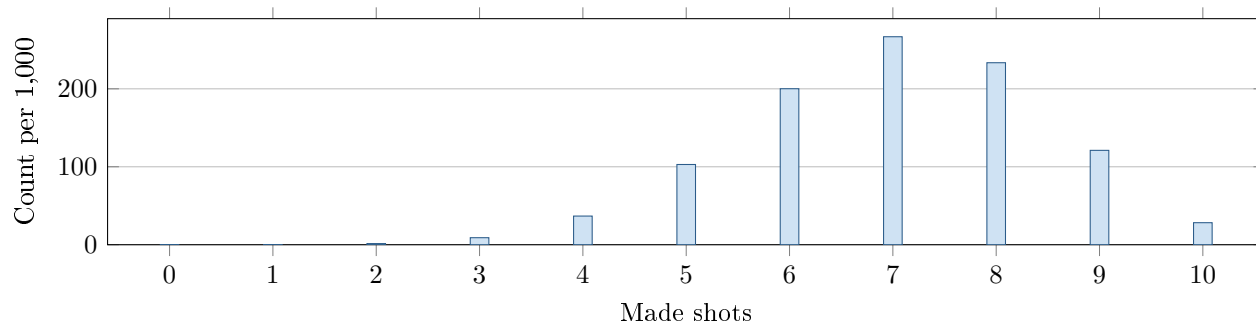
[STAR] THE PROBLEM — annotated

Suppose a basketball player has made 70 percent of her free throws over a long career. Tonight she makes 3 of 10 attempts. Let X be the number she would make in 10 attempts if her chance on each shot were still 0.70. The graph shows the binomial model as expected frequencies per 1,000 sets of 10 shots.

Claim A: “She is in a slump—making only 3 of 10 never happens at her career rate.”

Claim B: “With only 10 shots, 3 makes is ordinary chance variation.”

Under this model, technology gives $P(X \leq 3) = 0.0106$ for 3 or fewer makes.



FIRST TAKE — one sentence before you work the parts

Which claim, if either, do you trust more? Cite one detail from the result or graph.

Key: Not graded. Preserve the possibility of neither claim; both statements intentionally overreach in different directions.

[BOOK] BINS BEFORE BINOMIAL (keep on the page)

A binomial count uses Binary outcomes, Independent trials, a fixed Number of trials, and the Same success chance. Define X , n , and p . For exactly x successes, $P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$. A cumulative probability adds the included counts. A small probability means unusual under the model, not impossible; it does not identify a cause.

DOK 1 (a) State what X counts, and give n and p .

Key: X counts makes in 10 attempts; $n = 10$ and $p = 0.70$.

DOK 2 (b) Calculate $P(X = 3)$ using the binomial formula. Explain how this event differs from $X \leq 3$.

Key: $P(X = 3) = \binom{10}{3} (0.70)^3 (0.30)^7 = 0.009001692 \approx 0.0090$. Exactly 3 includes one count; $X \leq 3$ includes 0, 1, 2, and 3 makes.

DOK 3 (c) Is either claim fully supported? Use the given chance of 3 or fewer makes to explain what each claim gets wrong. Name one reason independence or the same success chance might fail, and one kind of new evidence that would help. Write 3–4 sentences.

Key: Neither claim is fully supported: $P(X \leq 3) = 0.0106$ makes this unusual under the model, contrary to B, but possible, contrary to A. One unusual set does not prove that the player’s rate changed. Fatigue or injury could make a constant 0.70 chance implausible. More attempts under comparable conditions would help: a sustained rate below 70% would strengthen the slump explanation.